Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/17/2012
PI Name:	McKinley, Gareth Ph.D.		
Project Title:	The Effects of Shear History on the Extensional Rheology of Complex Fluids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/08/2007	**End Date:**	03/07/2011
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-120 ; STS-123 ; STS-126 ; STS-128 ISS NOTE: original end date was 3/7/2008; extended to 3/7/2009 in 03/2008; extended to 03/07/2011 in 06/2009 (10/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Soulages, Johannes (Massachusetts Institute of Technology)		
Grant/Contract No.:	NNX07AE08G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The extensional viscosity is a fundamental physical property of all non-Newtonian materials, and cannot be determined from simple viscometric shear flow experiments. Constitutive equations for viscoelastic fluids such as dilute polymer solutions predict large changes in the extensional viscosity as the elongation rate is increased; however, the validity of these theories has only recently been confirmed due to paucity of experimental data that were available in extensional flows. To date, most quantitative rheological measurements in shear-free flows have only been possible for highly elastic or "stiff" materials such as polymer melts which can easily be elongated without sagging under a gravitational body force. In a review for the National Research Council on the future of Fluid Dynamics Research, Denn (1993) remarked that "...there are two major outstanding problems in rheological measurements. One is the measurement of extensional stresses of mobile fluids such as polymer solutions." Previous research on filament stretching rheometry supported by NASA in the P.I.'s lab has gone at least some way to addressing this situation. The perturbative effects of gravity and/or container boundaries that have interfered with almost all previous studies of extensional rheology in mobile fluids have been addressed in two ways; (i) by miniaturizing the apparatus or using neutral buoyancy tanks to minimize body forces, or (ii) by performing similar experiments in an extended microgravity environment. Using the instrumentation developed it has become possible to obtain accurate measurements of the extensional viscosity for more 'mobile' fluids such as dilute and concentrated polymer solutions, gels, suspensions and liquid crystalline materials. This rheological data is important in validating or extending constitutive models that aim to describe the response of complex fluids in arbitrary deformation fields. Such models will ultimately allow designers of both space- and ground-based material processes to use numerical simulations for a priori predictions of the properties of new materials that involve complex two- and three-dimensional fluid flows. The objectives of the proposed research are to (i) adapt an existing filament stretching device developed at MIT in order to enable extensional measurements of a wider range of complex fluids including worm-like micellar fluids and associative polymer gels; (ii) To combine these mechanical measurements with a microstructural probe of the evolution of the molecular orientation of the polymer chains using flow-induced birefringence; (iii) to explore, for the first time, the effects of a pre-shear history on the transient extensional viscosity in a uniaxial stretching flow for dilute polymer solutions together with the subsequent relaxation behavior after cessation of extensional deformation; (iv) to investigate the viscoelastic free-surface adhesion instabilities and the process of fibril formation in viscoelastic fluids which contribute significantly to the work of adhesion that must be input to separate two rigid planar substrates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The SHERE Experiment is an ISS glovebox experiment designed to study the effect of preshear on the transient evolution of the viscoelastic tensile stresses for monodisperse dilute polymer solutions together with the subsequent fluid relaxation behavior after cessation of the extensional deformation. The combination of both rotational shearing and extensional stretching flows is common in many polymer processing operations. Therefore, a thorough understanding of the shear-free flows of polymeric liquids is required in order to accurately predict their extensional properties. This research has thus direct applications in many industrial processes such as film coating operations, stability and breakup of jets, enhanced oil recovery and turbulent drag reduction for advanced aircraft, boats, and submarines. In addition, understanding and quantifying the extensional rheology of complex fluids allows to improve and control earth-based manufacturing processes such as fiber micro/nanofabrication, atomization, electrospinning and to understand elastically-driven flow instabilities in microfluidics.
Task Progress:	[Editor's note 7/17/2012: No Task Book report received. Progress section and Bibliography compiled from PI's Final Technical Report.] The SHERE flight hardware and fluid samples were successfully launched on-board three different Space Shuttle Missions between October 2007 and November 2008. Between July 2008 and January 2009, more than 50 experiments allowed us to study the effects of a preshear history on the transient extensional viscosity of a dilute polymer solution in a uniaxial stretching flow in microgravity. Data processing protocols have been developed and refined in order to make various corrections for anomalous readings, particularly from the force transducer. These include a baseline offset, drift, a force step during the preshear stage, spike removal and smoothing of the data during stretching. Additionally we have compiled code that analyzes video data of the stretching fluid filaments using an edge detection filter to determine the mid-plane filament radius, Rmid. Analysis of the experimental data has revealed very interesting elastic instabilities on the free surface of the fluid sample as well as the formation of beads-on-a-string structures in the absence of gravitational sagging. To our knowledge, the development and evolution of such phenomena upon cessation of elongation have not yet been described. The transient apparent extensional viscosity and Trouton ratio has been determined for test points where such analysis was possible in the absence of rotational instabilities. At low extension rates no clear trends were observed in the value of the Trouton ratio as the preshear rate was varied, however this is possibly due to the significant noise in the force measurements. At higher extension rates, where the measured forces are greater than the noise, the Trouton ratio tends to decrease with an increasing degree of preshear, consistent with the results of Anna and McKinley (Rheologica Acta, Vol. 47(8), pp. 841, 2008). A general trend that was observable across all test points was a reduction in Trouton ratio with increasing extension rate for fixed preshear conditions. This effect diminished when no preshear was imposed. We have also developed Matlab code to determine the transient extensional viscosity, Trouton ratio and accumulated Hencky strain during filament thinning following the cessation of the 29 uniaxial stretch. The results suggest that this phase of the experiment is not significantly affected by the initial preshear conditions. The complete analysis of the SHERE data has enabled us to better understand the effects of rotational preshearing on the stress/strain response of a polymeric liquid being stretched in a microgravity environment. In the near future we hope to obtain new data from the SHERE II test fluid. The SHERE II project will utilize the existing SHERE hardware aboard the ISS to study the influence of rotational preshear on the extensional rheological properties of filled polymeric fluids. The current and future measurements will provide an idealized data set for theoretical model comparison and serve as a reference standard for ground-based extensional rheometry. They will also shed light on ongoing developments of "containerless processing" operations for the future construction of a permanent lunar base.

Bibliography Type:	Description: (Last Updated: 09/04/2025)